

Emile Gaudinot

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EDUCATION

- **Technical University of Berlin** Berlin, Germany
Applied Mathematics, Statistics, Data Science (MSc), GPA: 4.0 2022 – 2023
- **École Centrale de Nantes** Nantes, France
Engineering Degree (MSc), GPA: 4.0 2020 – 2022

EXPERIENCE

- **Sia** Paris, France
Experienced Data Science Consultant Feb. 2026 – Present (9 months)
 - **Electrical Network RL Optimization (Major Energy Provider)**: Using **Reinforcement Learning (PPO)** and a **GNN**, optimized the critical operations on the Electricity Network. **+500%** speed vs. current solvers.
 - **Revenue Forecast (Luxury Client)**: Trained and deployed in production a **Holt-Winters** model on **GCP Vertex AI**. Predicted daily sales over 12 months for each of the 600 stores in the world. **-17%** WMAPE vs. sales plans.
 - **Video Moderation System (Media Client)**: Automated the blurring of alcohol, cigarettes and firearms with **YOLO26/RF-DETR** models on **AWS SageMaker**. **React** dashboard generated from a **Langflow** agentic pipeline. App fully deployed on **Databricks**.
 - **Penguin Challenge (World-Wide Medical Challenge)**: Automated the segmentation of bone fragments on pelvis fracture **CT-scans** with 2 **nnUNet**. Reduction planning with **PointNet** + **Transformer**. Top 100 world.
- **Fraunhofer - Heinrich-Herz-Institute** Berlin, Germany
Experienced Data Scientist Mar. 2025 – Feb. 2026 (1 year)
 - **Perceived Distance Prediction from Brain Signals**: Developed an innovative AI pipeline using **MNE-Python** with (1) an **SSD** and a **SPoC regression** for distance, and (2) a **ROC classifier** for personal space intrusion detection ($F1 = 0.81$). Extension of the pipeline for **real-time** using **LSL** ($F1 = 0.65$) and deployed with **Databricks Workflows**. Published the results in a [research paper](#).
- **Charité** Berlin, Germany
Data Scientist Mar. 2023 – Feb. 2025 (2 years)
 - **Computer Vision for Early Detection of Coronary Artery Disease**: Segmented arteries from 3D heart images with **U-Net**, **U-net Transformer** and **MedSAM**, on **MONAI** ($IoU = 0.79$). Implemented the pipeline for artery graph extraction with **scikit-image** (**CCA**, **skeletonization**, etc). Monitoring with **MLflow**. Released in production with **Databricks**, leading to **+60%** correct CAD diagnosis. Published the results in a [research paper](#).
 - **Intelligent Computer Security**: (1) Implemented a malware detector for .pdf and .docx using **Bag-of-Words**, **PCA** and **XGBoost** ($F1 = 0.98$) (2) Automated a network intrusion detector using **Isolation Forests** and **LOF scores** (**-120%** false positives) (3) Robustified models to **Adversarial Attacks** using **i-FGSM**.

PROJECTS

- **Winner of BIFOLD's Cybersecurity Hackathon (Oct. 2023 – Mar. 2024)**: Trained and fine-tuned ML models on 4 cybersecurity challenges. Price Ceremony (1st Price/52) at the German Ministry of Research, Technology and Space.
- **Local LLM - Complete app (Apr. 2025 – Nov. 2025)**: Developed and optimized a custom [ChatBot](#) application which runs locally. Possibility to upload documents (img, pdf, docx, pptx) thanks to an OCR + text extraction pipeline.

SKILLS

- **LLMs**: Mistral API, Ollama, Light local models (ministral-3.8b, gemma4-e4b, llama3.1-8b)
- **Deep Learning**: RL (ActorCritic, PPO), GNN (GAT, GCN, GraphSAGE), Time series (RNN, LSTM), Vision (nnU-Net, UNETR, MedSAM, YOLO26, RF-DETR)
- **Tools**: Production (AWS, GCP, Databricks, Docker, MLflow, Git), Python (PyTorch, Scikit-learn, Pandas), HuggingFace
- **Programming Languages**: Python, SQL, Bash, C++, R
- **Spoken Languages**: French (native), English (C1 - TOEFL 104/120), German (C1 - ZEMS 92/100)

Certifications: Databricks Certified Machine Learning Associate, ANSSI SecNum: Cybersecurity & Digital Tool Protection